Data Insights for Optimizing Automobile Sales Strategies Report

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About the Data and Its Dictionary

Data Overview:
The dataset provides comprehensive details about individuals and their vehicle purchases, including personal information, economic information, and specifics related to the vehicles.

Data Dictionary:

- **Age**: Age of the individual
- **Gender**: Gender of the individual
- **Profession**: Profession of the individual (Salaried/Business)
- **Marital_status**: Marital status of the individual (Married/Single)
- **Education**: Education level of the individual (Graduate/Post Graduate)
- **No_of_Dependents**: Number of dependents
- **Personal_loan**: Whether the individual has a personal loan (Yes/No)
- **House_loan**: Whether the individual has a house loan (Yes/No)
- **Partner_working**: Whether the individual's partner is working (Yes/No)
- **Salary**: Salary of the individual
- **Partner_salary**: Salary of the individual's partner
- **Total_salary**: Total household salary
- **Price**: Price of the vehicle purchased
- **Make**: Make of the vehicle (SUV/Sedan/Hatchback)

Tables

Missing Values Count and Percentage:

	Count	Percentage
Gender	53	3.352309
Partner_salary	106	6.704617

Misspelled Gender Value Counts:

Gender	Value Count
Male	1252
Female	327
Femal	1
Femle	1

Corrected Gender Value Counts:

Gender	Value Count
Male	1252
Female	329

Methodology

1. Data Collection

- The dataset used for this analysis was obtained from the austo_automobile.csv file.
- The dataset provides comprehensive details about individuals and their vehicle purchases, including personal information, economic information, and specifics related to the vehicles.

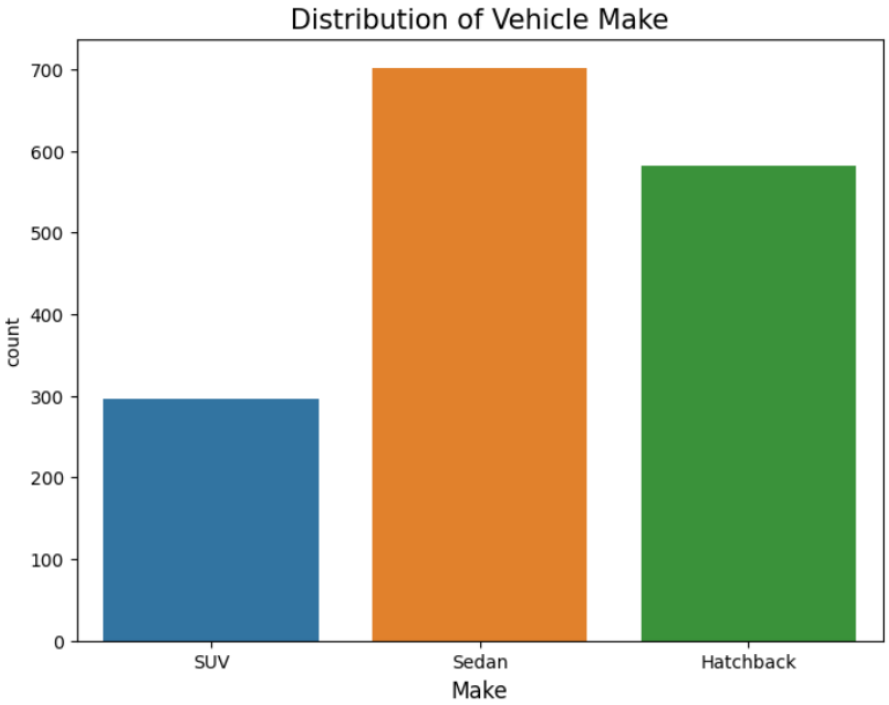
2. Data Cleaning

- **Handling Missing Values**:
  - Missing values in the 'Partner_salary' column were filled with 0 and converted to integer type.
  - Missing values in the 'Gender' column were filled with the mode value.
- **Correcting Errors**:
  - Misspelled values in the 'Gender' column ('Femal' and 'Femle') were corrected to 'Female'.
  - **Updating 'Partner_salary'**:
    - The 'Partner_salary' column was updated where 'Partner_working' is 'Yes' and 'Partner_salary' is 0. The new value was calculated as the difference between 'Total_salary' and 'Salary'.
- **Handling Outliers**
  - Outliers were handled by replacing them with the median values for the numeric columns.
  - It has improved the overall data quality, making it more reliable for analysis and modeling.

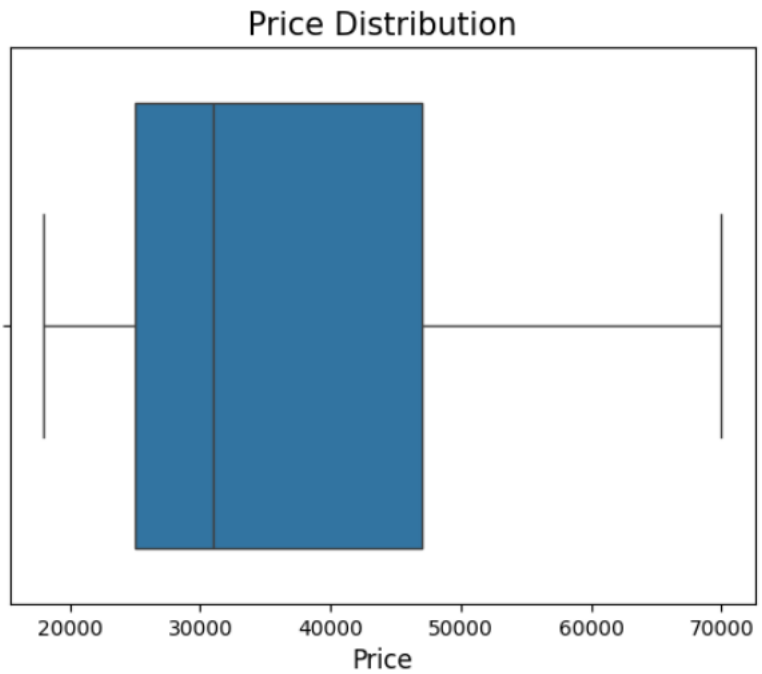
3. Exploratory Data Analysis (EDA)

- **Descriptive Statistics**: Calculated basic statistics such as mean, median, and mode for numerical columns.
- **Value Counts**: Analyzed the distribution of categorical variables using value counts.
- **Visualizations:** Utilized Univariate and Bivariate Analysis

Univariate Analysis

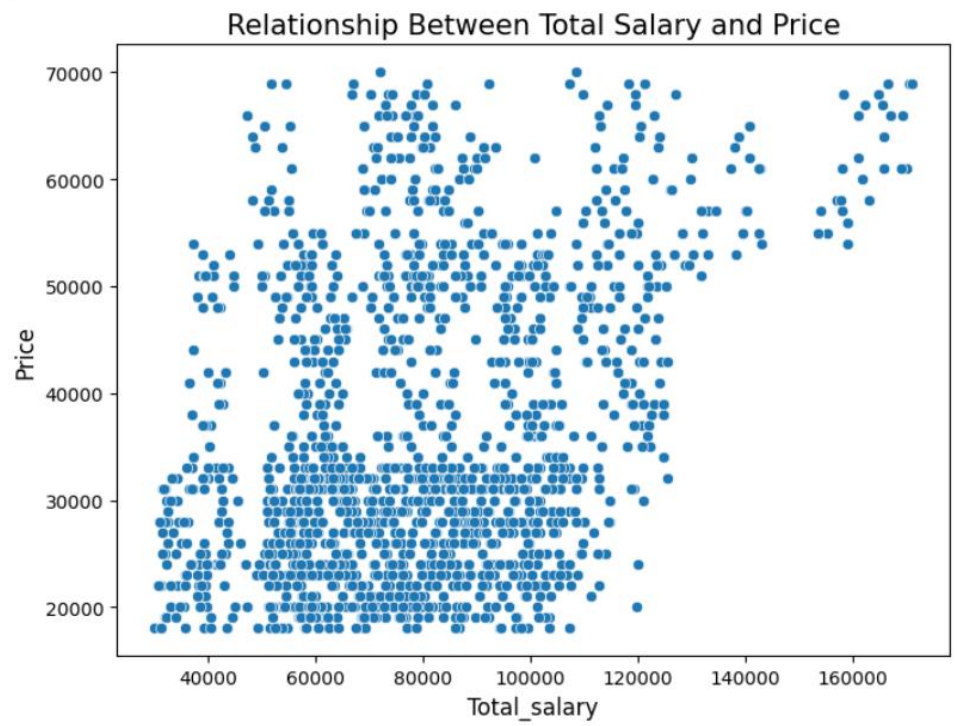


Sedans are the most common vehicle make in the dataset, followed by Hatchbacks and SUVs.



Displays a right-skewed distribution, with most vehicles priced between ₹20,000 and ₹40,000.

Bivariate Analysis



The scatter plot shows a positive correlation between total salary and vehicle price.

This plot highlights gender-based differences in salary progression with age.

Questions and its Answers

1. Do men tend to prefer SUVs more compared to women?

Output:

```
Gender
Female    58.249158
Male      41.750842
```

- **Observation**: The preference for SUVs among genders is as follows:
  - Female: 58.24%
  - Male: 41.80%
- **Insight**: Women tend to prefer SUVs more compared to men, as a higher percentage of SUV purchases are made by females

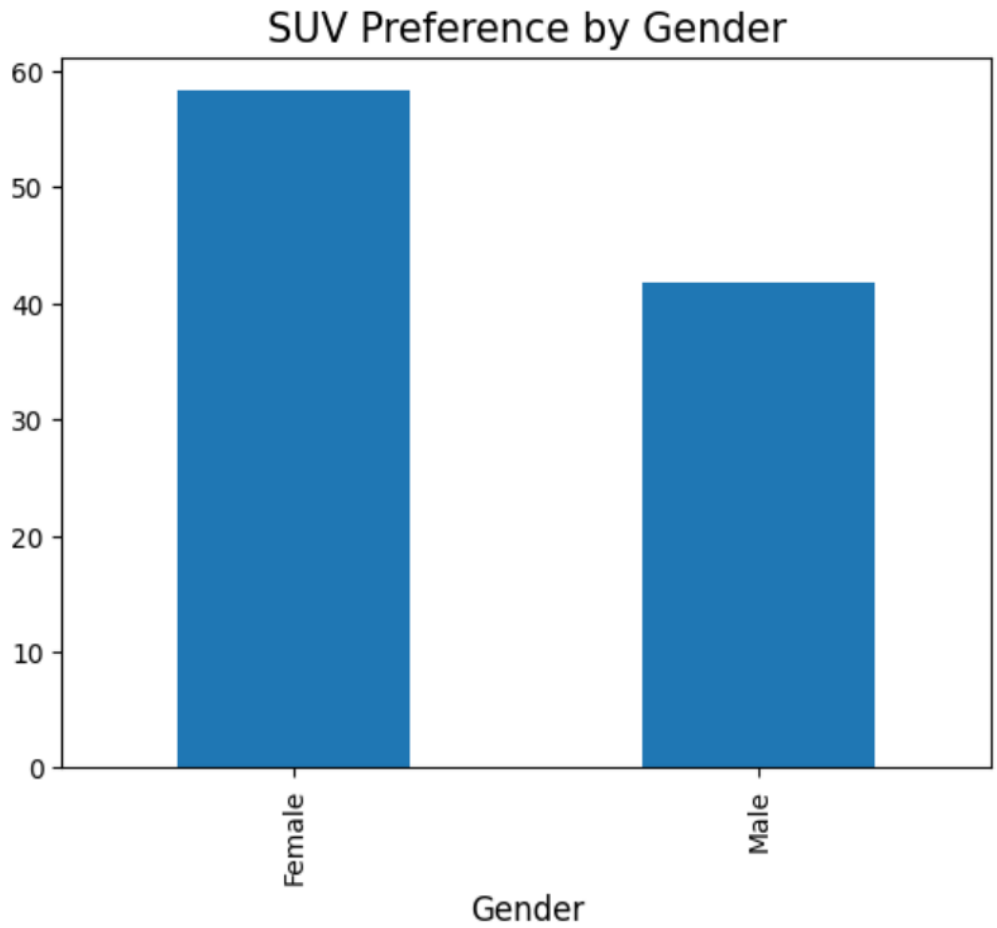


Figure (a)

2. What is the likelihood of a salaried person buying a Sedan?

Output:

```
Make
Sedan        44.196429
Hatchback    32.589286
SUV          23.214286
```

- **Observation**: The likelihood of a salaried person buying a Sedan is 44.20%.
- **Insight**: Nearly half of the salaried individuals prefer Sedans.

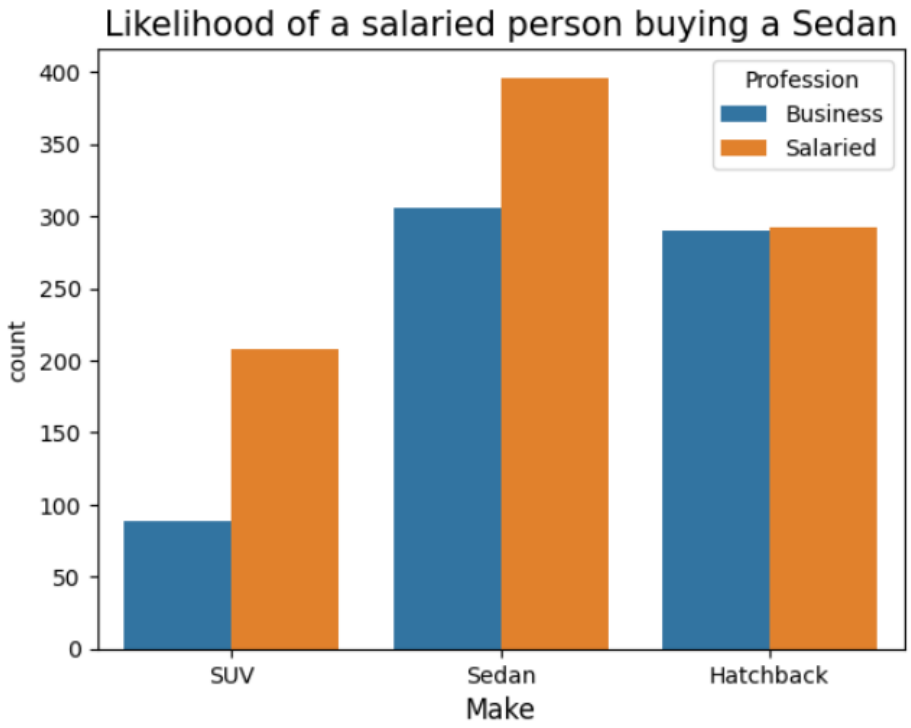


Figure (b)

3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?

Output:

```
Make
Sedan        45.386905
Hatchback    41.220238
SUV          13.392857
```

- **Observation**: The distribution of vehicle purchases among salaried males is as follows:
  - Sedan: 43.59%
  - Hatchback: 43.13%
  - SUV: 13.28%
- **Insight**: The data does not support Sheldon Cooper's claim. Salaried males are more likely to buy Sedans or Hatchbacks over SUVs.

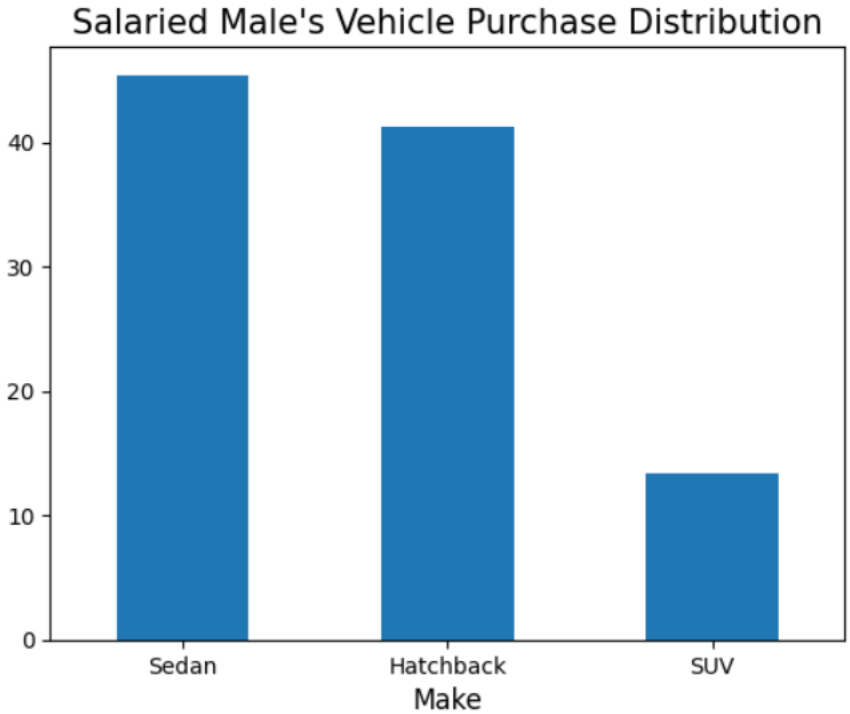


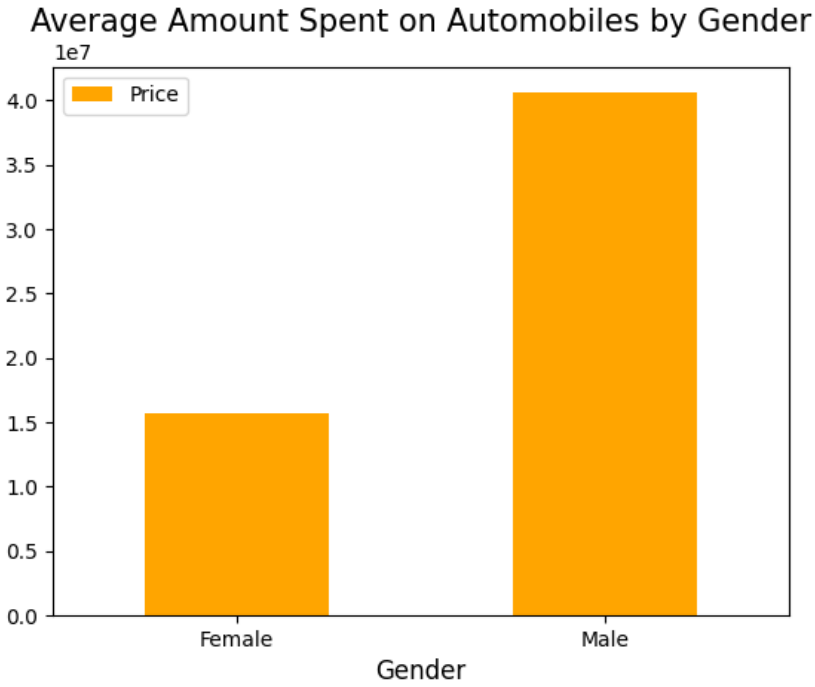
Figure (c)

4. How does the amount spent on purchasing automobiles vary by gender?

Output:

	Price
Gender	
Female	15695000
Male	40585000

- **Observation**: The average amount spent on purchasing automobiles by gender is as follows:
  - Female: 15,695,000
  - Male: 40,585,000
- **Insight**: Males tend to spend more on automobiles compared to females.



Figure(d)

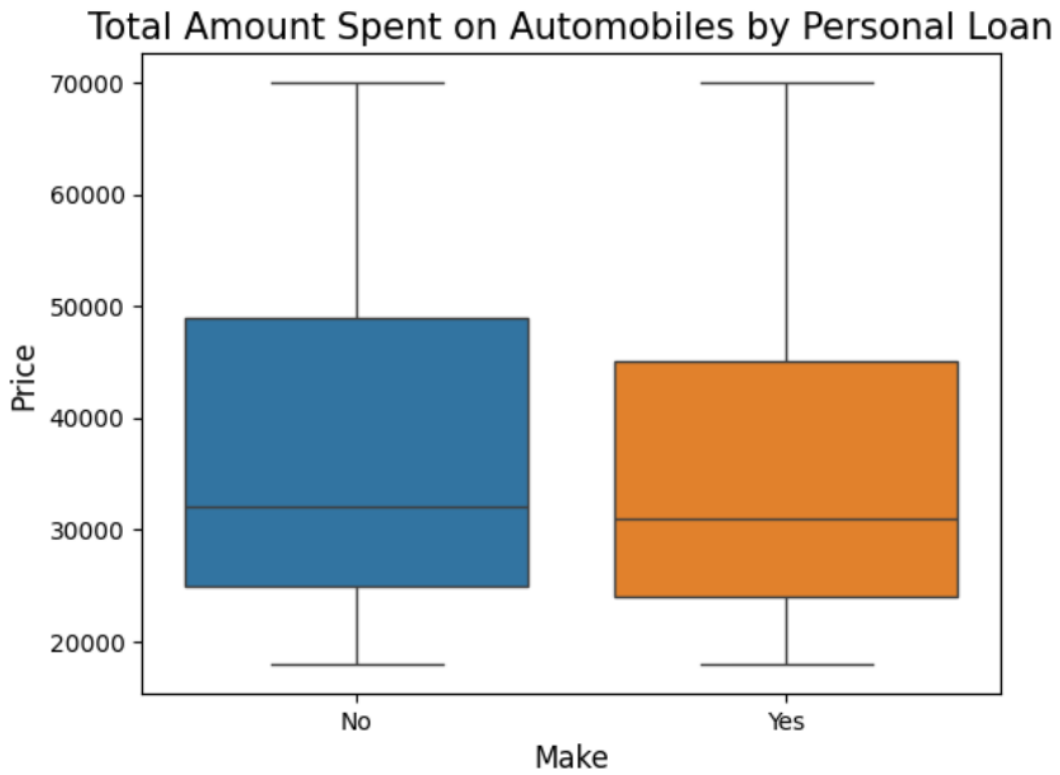
5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

Output:

```
                  Price
Personal_loan
No             28990000
Yes            27290000
```

- **Observation**: The total amount spent on purchasing automobiles by individuals who took a personal loan is 27,290,000.
- **Insight:** A substantial amount of money (27,290,000) was spent

on purchasing automobiles by individuals who took a personal loan.



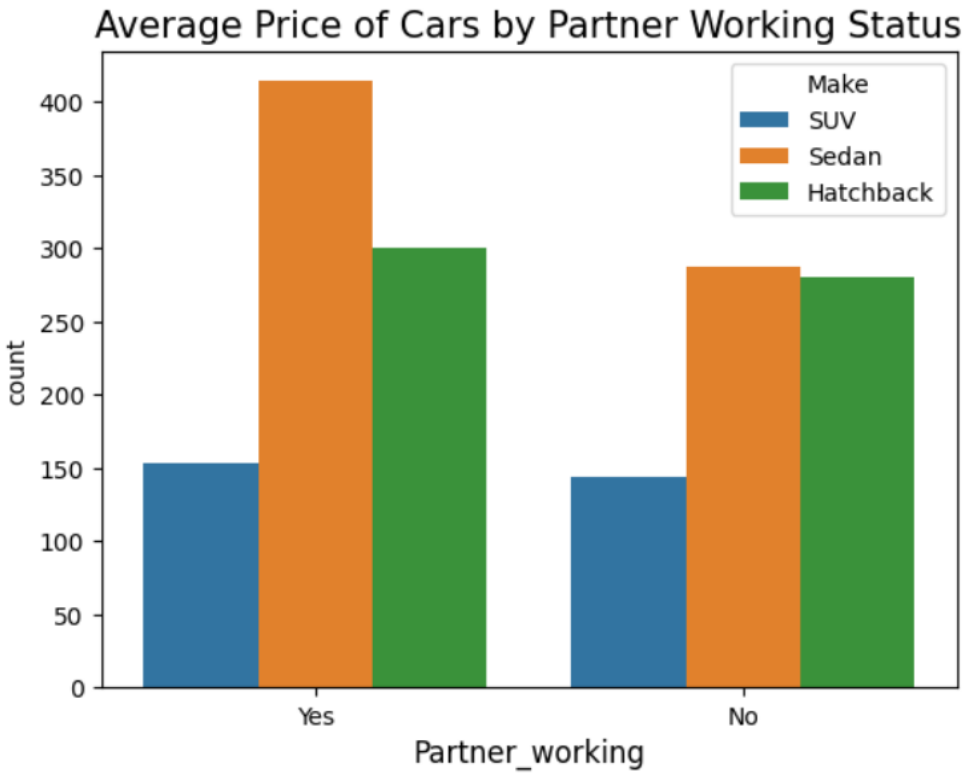
Figure(e)

6. How does having a working partner influence the purchase of higher-priced cars?

Output:

Partner_working	Price
No	36000.000000
Yes	35267.281106

- **Observation**: The average price of cars purchased by individuals with a working partner is ₹35,267.28, while the average price for those without a working partner is ₹36,000.
- **Insight**: Having a working partner does not significantly influence the purchase of higher-priced cars.



Figure(f)

Actionable Insights

1. **Gender Preferences:** Women tend to prefer SUVs more compared to men.
2. **Salaried Individuals:** Nearly half of the salaried individuals prefer Sedans.
3. **Vehicle Purchase Distribution:** Salaried males are more likely to buy Sedans or Hatchbacks over SUVs.
4. **Spending by Gender:** Males tend to spend more on automobiles compared to females.
5. **Personal Loans:** A significant amount of money was spent by individuals who took personal loans.
6. **Partner Working Status:** Having a working partner does not significantly influence the purchase of higher-priced cars.

Business Recommendations

1. **Target Marketing Campaigns**: Focus marketing efforts on men for SUVs.
2. **Special Promotions**: Offer special promotions for Sedans to salaried individuals.
3. **Promote Sedans and Hatchbacks**: Focus on promoting Sedans and Hatchbacks to salaried males.
4. **Premium Vehicle Models**: Develop premium vehicle models targeting female customers.
5. **Financing Options**: Provide attractive financing options for individuals taking personal loans.
6. **Exclusive Events and Promotions:** Organize exclusive events or promotions for families with two incomes.